Mohammed Kord

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Research interests: machine learning, information retrieval, ML systems and computer vision.

Education

Palestine Polytechnic University, Hebron *Sep 2025 – Expected 2029*
B.Sc. in Computer Engineering. GPA: 96.1/100, ranked 1st in cohort after the first year. Currently in year two.

Self-study: PyTorch, neural networks, CNNs, embeddings and retrieval. Calculus and selected topics from real and complex analysis, linear algebra and abstract algebra.

Alhajj Radwan & Alhajah Suad Abu Sneineh Private Secondary School for Boys *2025*
Tawjihi (General Secondary Education Certificate), scientific stream: 99.4/100, joint 10th in the Middle Hebron Education Directorate (top score 99.7).

Projects

RAG service for STEM textbooks *C#, ASP.NET Core, MinerU, Ollama, Qdrant*
github.com/MohammadAlkurd/RAG

- Built a self-hosted service for STEM textbooks that converts PDFs to Markdown (including LaTeX formulas) with MinerU, splits the content by headings, generates embeddings locally with Ollama and stores them in Qdrant. A query script retrieves relevant chunks and passes them to a local LLM, which answers with section citations.
- Added an upload API with validation and a bounded background job queue, with page-by-page progress reporting and SHA-256 hashing to avoid processing the same book twice. Implemented GPU hand-over so OCR and the LLM can both run on an 8 GB GPU.
- Tested on a 789-page calculus textbook: OCR in about 13 minutes, 1,322 chunks. On 60 real Math StackExchange questions, a chunk with the full answer was in the top 5 for 63% (87% counting partial answers). Further info in README.

Hand Gestures To Action *Python, PyTorch, MediaPipe, PyQt6*
github.com/MohammadAlkurd/Hand-Gestures-To-Action

- Built a desktop app that captures custom hand gestures from a webcam, extracts hand landmarks with MediaPipe, trains a small MLP classifier, and runs live inference to trigger keypresses or scripts.
- First tried a CNN on raw frames, which gave low accuracy, so I switched to MediaPipe hand landmarks and an MLP. Added mirroring augmentation so it works for both hands. New gestures can be added by recording a short video and retraining the classifier.

Ragdoll reinforcement learning *Godot 4, C#, Python*
github.com/MohammadAlkurd/RL_godot

- Trained a ragdoll to walk with PPO using the Godot RL Agents plugin. Found that weighting forward movement above staying upright trained faster, and that too low an upright weight made the agent crawl.

Games *Godot, Unity, C#*
mohammadalkurddev.itch.io

- Published three games on itch.io, including a two-player Quoridor implementation.

Awards & Honours

2025–present	Full merit scholarship (100%), Palestine Polytechnic University — awarded each semester for a semester average of 95/100 or higher.
2026	11th place (team of three), Palestinian Collegiate Programming Contest — national ACPC qualifier.

Activities

- Member of PPU's Competitive Programming Club (Code Academy), taking part in practice competitions and problem-solving sessions.

Experience

Freelance Android Developer, Upwork *2026*

- Built and delivered two Android applications in Kotlin for an Upwork client, both rated 5/5.

Skills

Programming	Python, C#, C++, Kotlin, SQL
ML	PyTorch (MLP, CNN), MediaPipe, reinforcement learning (PPO), embeddings and vector retrieval (Qdrant, Ollama)
Backend	ASP.NET Core, Entity Framework Core, REST APIs, background job queues
Tools	Git, Linux, Docker, GitHub Actions
Languages	Arabic (native), English (fluent)